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(54) **HIGH ELECTRON MOBILITY TRANSISTOR DEVICE AND MANUFACTURING METHOD THEREOF**

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See application file for complete search history.

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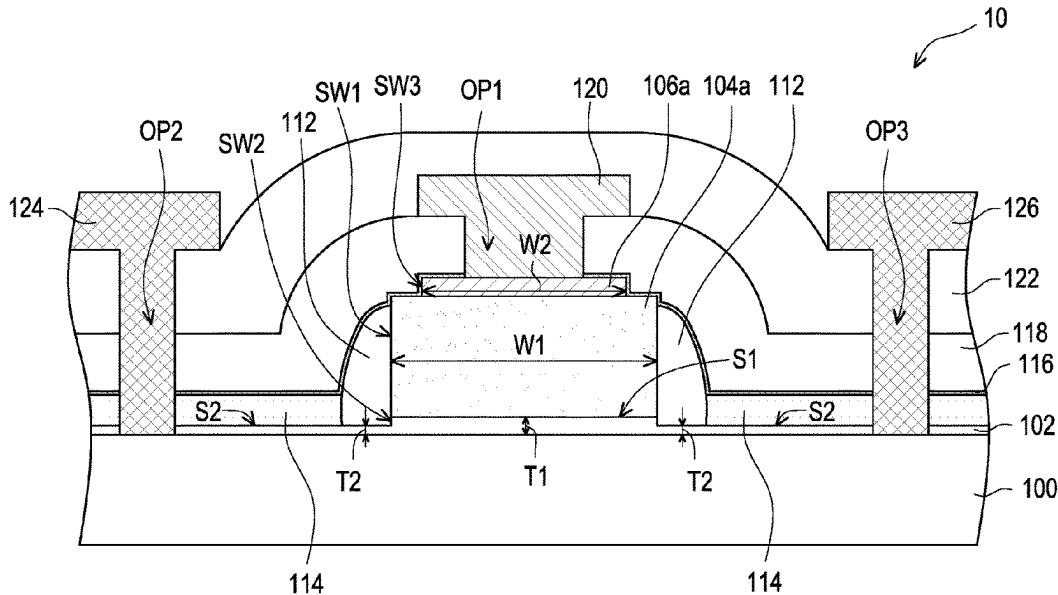
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(57) **ABSTRACT**

A high electron mobility transistor device including a channel layer, a first barrier layer, and a P-type gallium nitride layer is provided. The first barrier layer is disposed on the channel layer. The P-type gallium nitride layer is disposed on the first barrier layer. The first thickness of the first barrier layer located directly under the P-type gallium nitride layer is greater than the second thickness of the first barrier layer located on two sides of the P-type gallium nitride layer.

14 Claims, 11 Drawing Sheets



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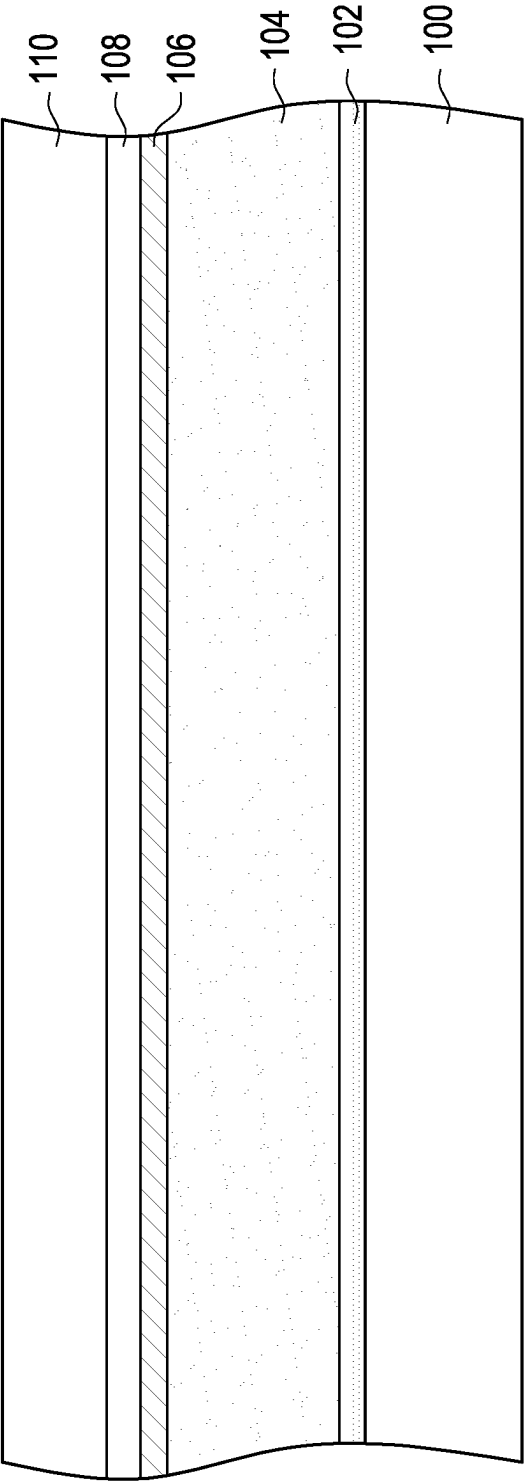


FIG. 1A

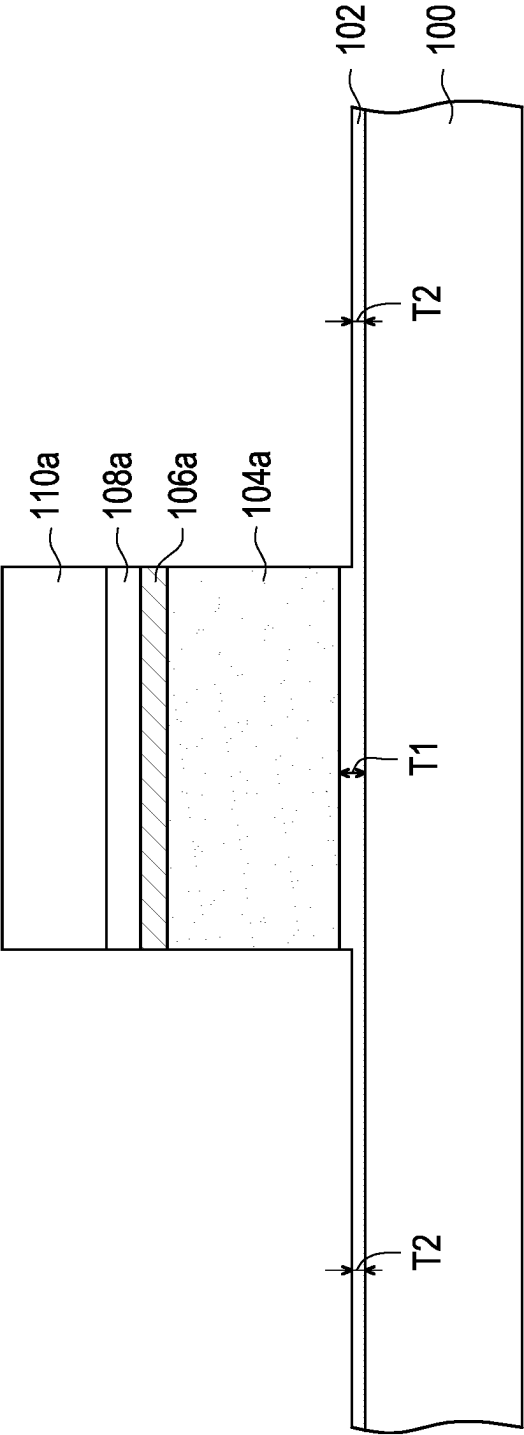


FIG. 1B

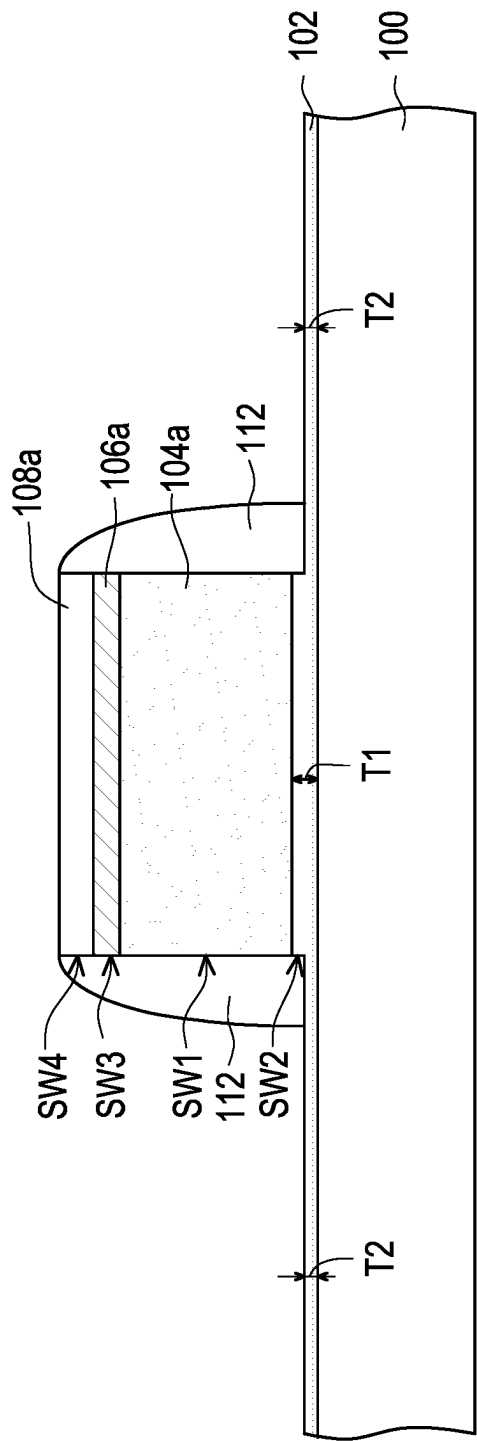


FIG. 1C

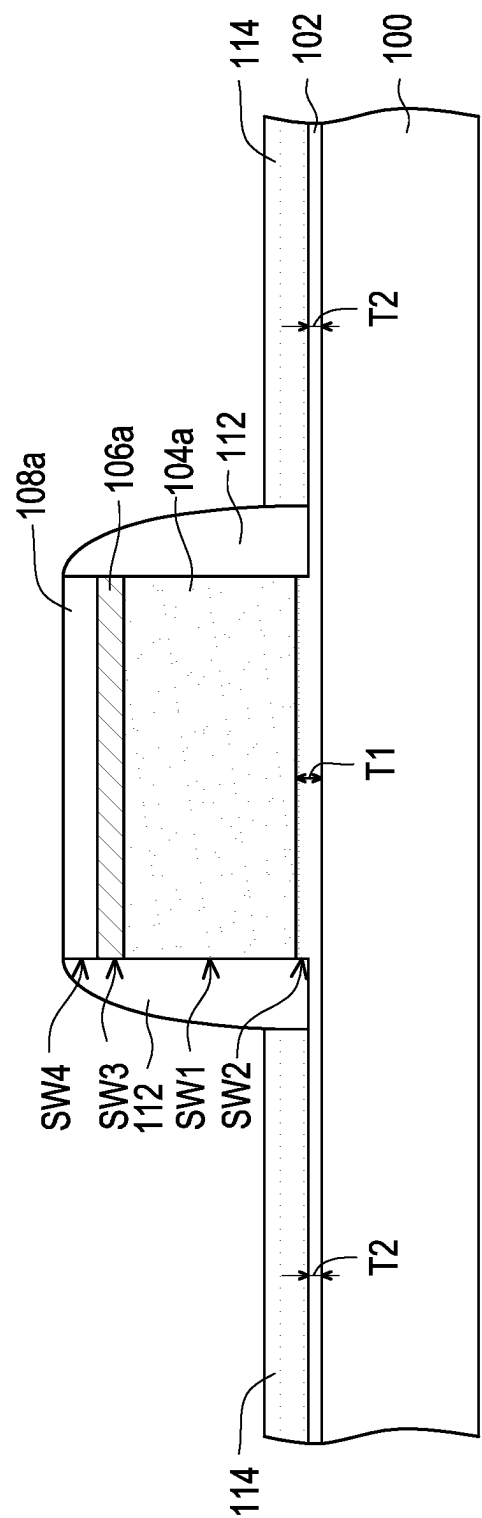


FIG. 1D

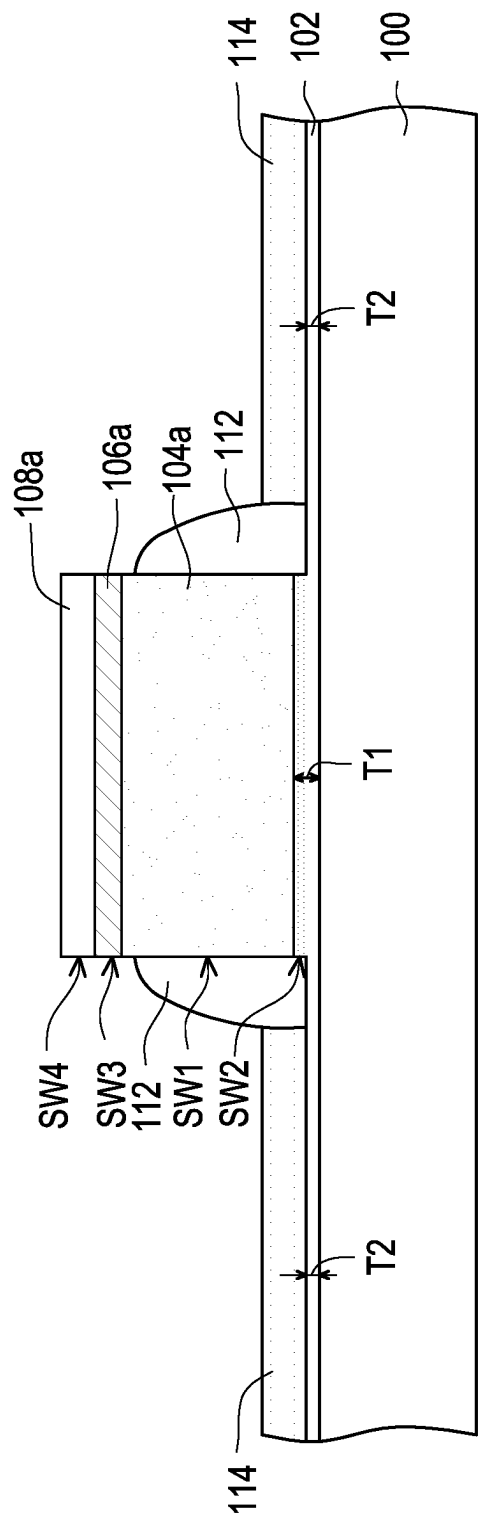


FIG. 1E

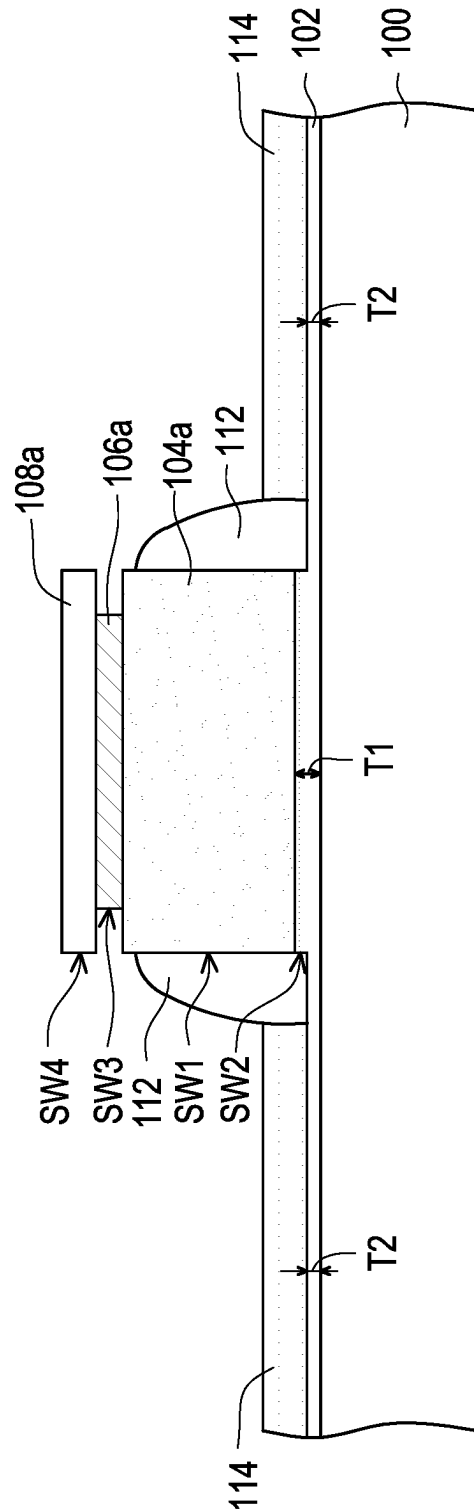


FIG. 1F

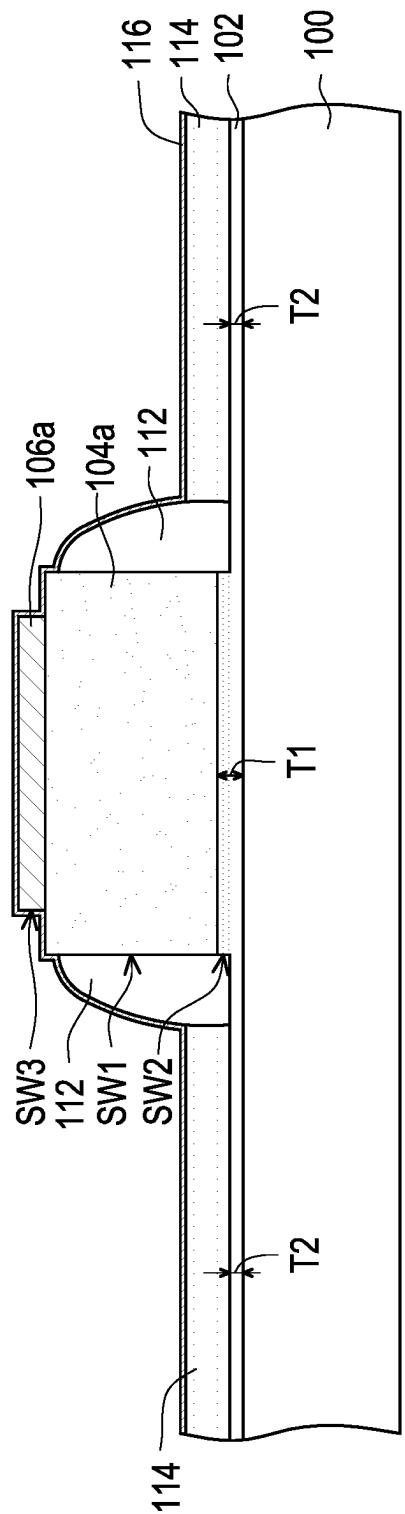


FIG. 1G

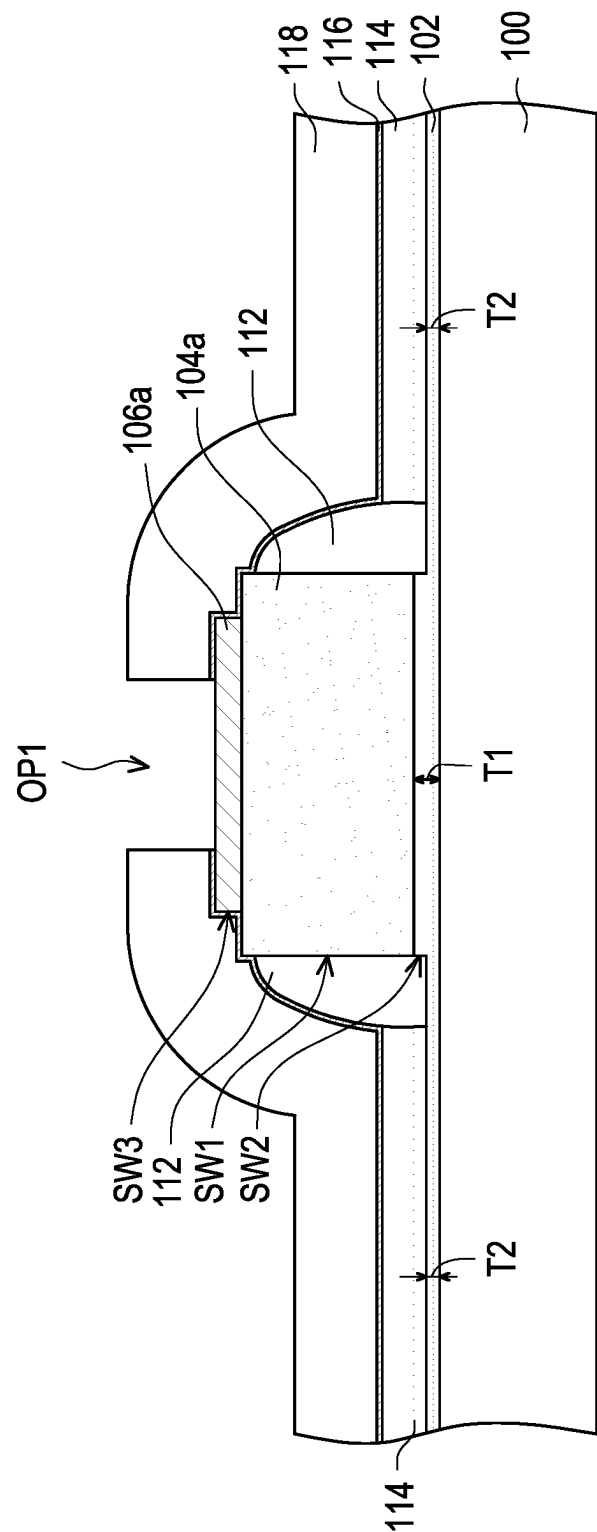


FIG. 1H

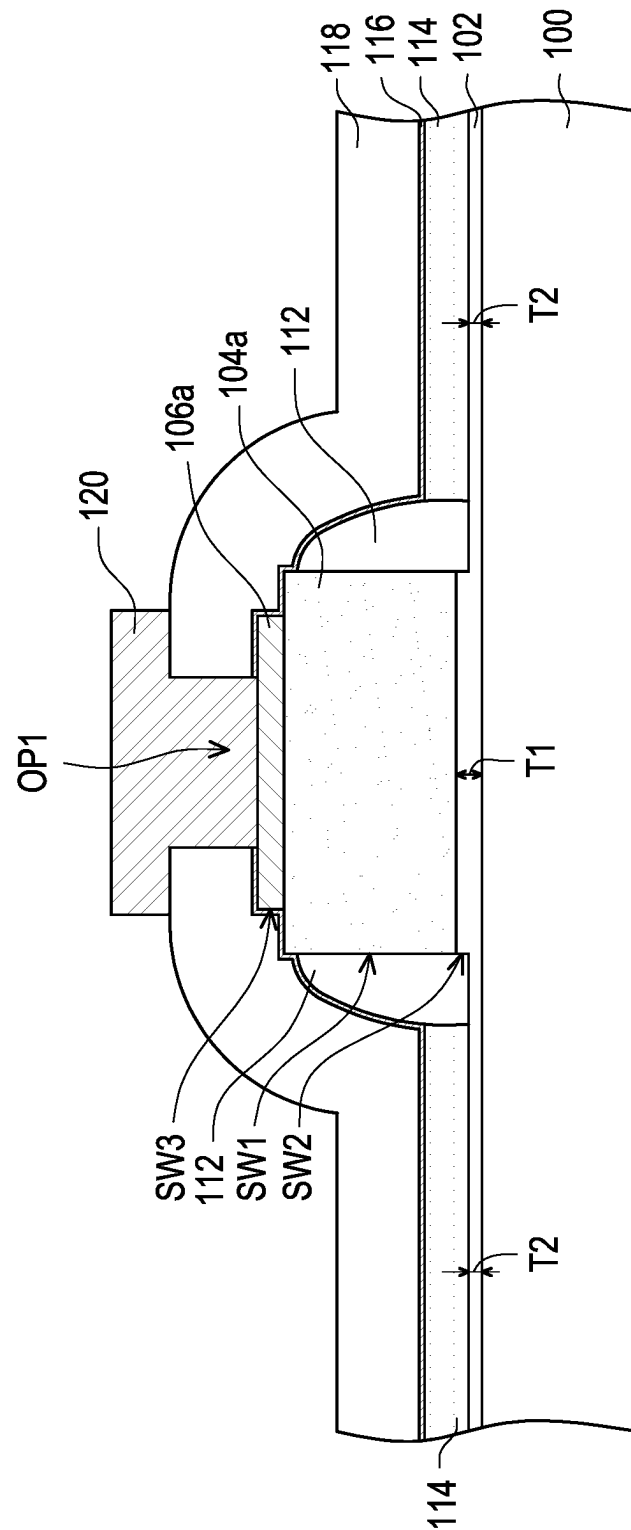


FIG. 11

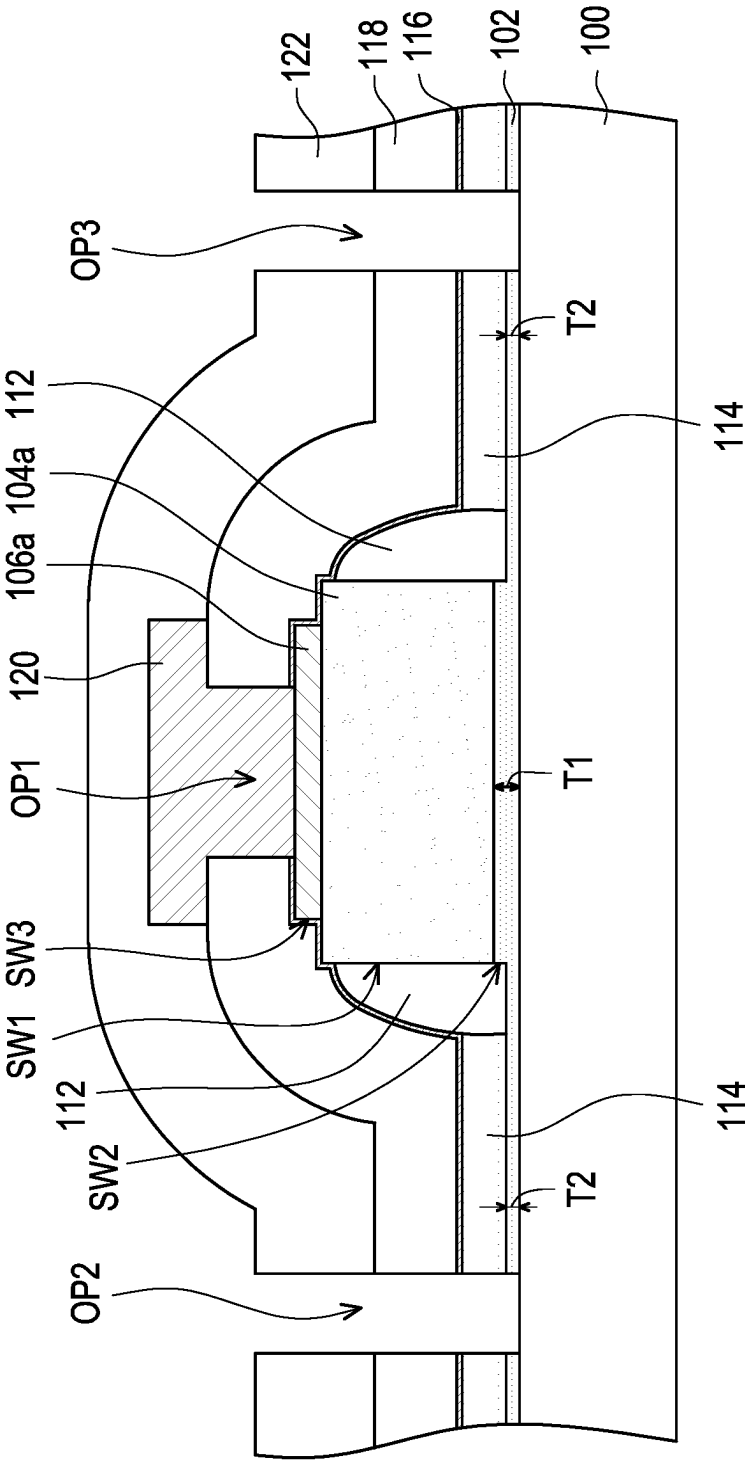


FIG. 1J

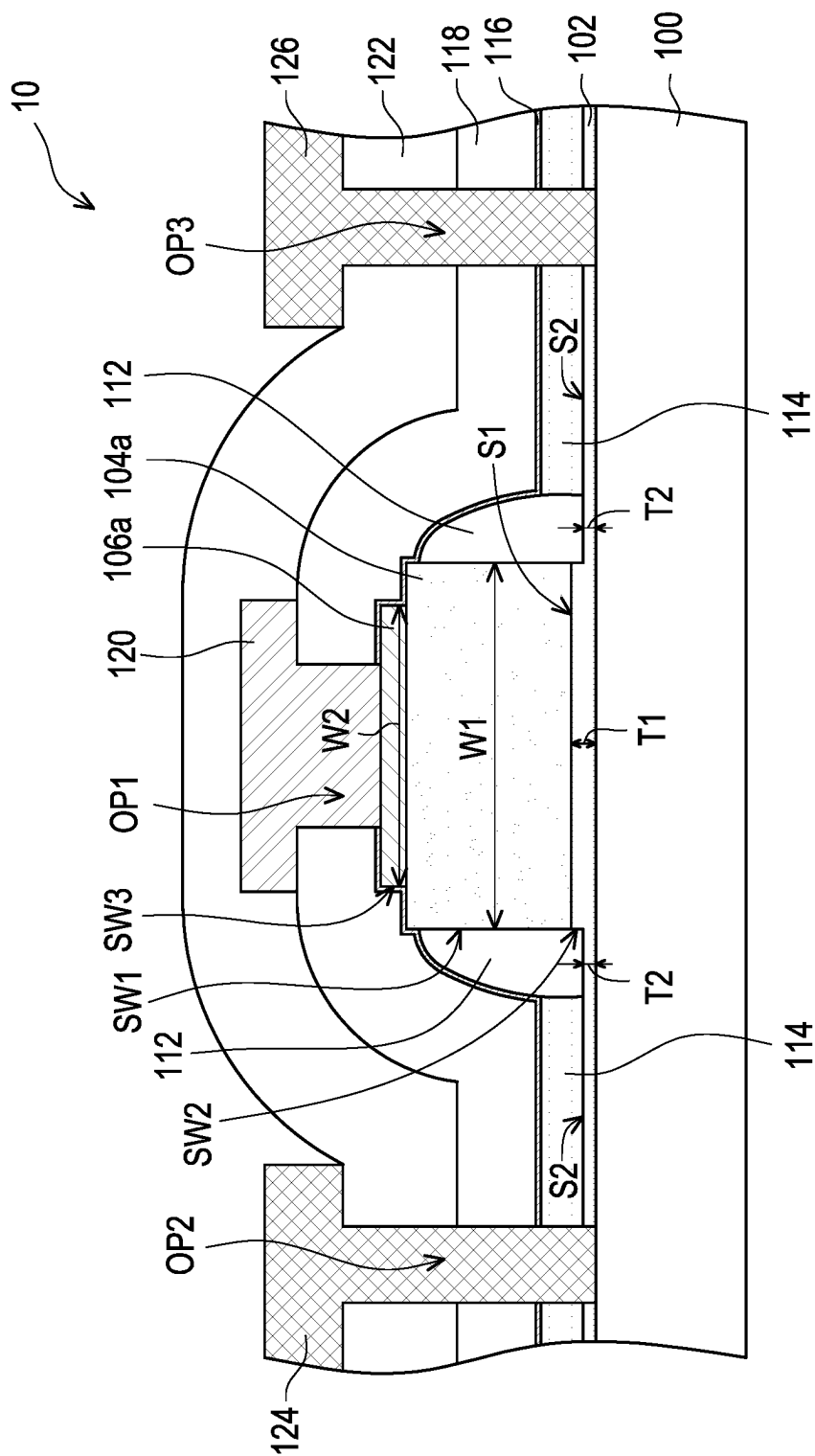


FIG. 1K

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HIGH ELECTRON MOBILITY TRANSISTOR DEVICE AND MANUFACTURING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of Taiwan application serial no. 111133309, filed on Sep. 2, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

The invention relates to a semiconductor device and a manufacturing method thereof, and particularly relates to a high electron mobility transistor (HEMT) device and a manufacturing method thereof.

Description of Related Art

The HEMT is a field effect transistor and can have high breakdown voltage and high reliability. However, a large electric field is often generated at the gate edge of the HEMT, which increases the leakage current of the HEMT, thereby reducing the reliability of the HEMT.

SUMMARY

The invention provides a HEMT device and a manufacturing method thereof, which can improve the reliability of the HEMT device.

The invention provides a HEMT device, which includes a channel layer, a first barrier layer, and a P-type gallium nitride (P-type GaN) layer. The first barrier layer is disposed on the channel layer. The P-type gallium nitride layer is disposed on the first barrier layer. The first thickness of the first barrier layer located directly under the P-type gallium nitride layer is greater than the second thickness of the first barrier layer located on two sides of the P-type gallium nitride layer.

According to an embodiment of the invention, in the HEMT device, the first thickness may be 14 nm (nanometer) to 24 nm.

According to an embodiment of the invention, in the HEMT device, the second thickness may be 2 nm to 22 nm.

According to an embodiment of the invention, in the HEMT device, the first upper surface of the first barrier layer located directly under the P-type gallium nitride layer may be higher than the second upper surface of the first barrier layer located on two sides of the P-type gallium nitride layer.

According to an embodiment of the invention, in the HEMT device, the material of the first barrier layer is, for example, aluminum gallium nitride (AlGaIn).

According to an embodiment of the invention, in the HEMT device, the aluminum content in the aluminum gallium nitride may be 5 atomic % to 50 atomic %.

According to an embodiment of the invention, the HEMT device may further include a second barrier layer. The second barrier layer is disposed on the first barrier layer. The second barrier layer and the P-type gallium nitride layer may be separated from each other.

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According to an embodiment of the invention, in the HEMT device, the material of the second barrier layer is, for example, aluminum gallium nitride.

According to an embodiment of the invention, in the HEMT device, the aluminum content in the aluminum gallium nitride may be 5 atomic % to 50 atomic %.

According to an embodiment of the invention, the HEMT device may further include a spacer. The spacer is disposed on the sidewall of the P-type gallium nitride layer and on the first barrier layer. The spacer may be located between the second barrier layer and the P-type gallium nitride layer.

According to an embodiment of the invention, in the HEMT device, the width of the spacer may be 0.1 μm (micrometer) to 0.5 μm .

According to an embodiment of the invention, the HEMT device may further include a passivation layer. The passivation layer is disposed on the P-type gallium nitride layer, the spacer, and the second barrier layer.

According to an embodiment of the invention, the HEMT device may further include a work function layer. The work function layer is disposed on the P-type gallium nitride layer.

According to an embodiment of the invention, in the HEMT device, the width of the work function layer may be smaller than the width of the P-type gallium nitride layer.

The invention provides a manufacturing method of a HEMT device, which includes the following steps. A channel layer is provided. A first barrier layer is formed on the channel layer. A P-type gallium nitride layer is formed on the first barrier layer. The first thickness of the first barrier layer located directly under the P-type gallium nitride layer is greater than the second thickness of the first barrier layer located on two sides of the P-type gallium nitride layer.

According to an embodiment of the invention, the manufacturing method of the HEMT device may further include the following step. A second barrier layer is formed on the first barrier layer. The second barrier layer and the P-type gallium nitride layer may be separated from each other.

According to an embodiment of the invention, the manufacturing method of the HEMT device may further include the following step. A spacer is formed on the sidewall of the P-type gallium nitride layer and on the first barrier layer. The spacer may be located between the second barrier layer and the P-type gallium nitride layer.

According to an embodiment of the invention, the manufacturing method of the HEMT device may further include the following step. A passivation layer is formed on the P-type gallium nitride layer, the spacer, and the second barrier layer.

According to an embodiment of the invention, the manufacturing method of the HEMT device may further include the following step. A work function layer is formed on the P-type gallium nitride layer.

According to an embodiment of the invention, the manufacturing method of the HEMT device may further include the following step. A lateral etching process is performed on the work function layer to reduce the width of the work function layer.

Based on the above description, in the HEMT device and the manufacturing method thereof according to the invention, the first thickness of the first barrier layer located directly under the P-type gallium nitride layer is greater than the second thickness of the first barrier layer located on two sides of the P-type gallium nitride layer. Therefore, the concentration of two-dimensional electron gas (2DEG) at the edge of the P-type gallium nitride layer can be reduced, thereby reducing the leakage current of the HEMT device and improving the reliability of the HEMT device.

In order to make the aforementioned and other objects, features and advantages of the invention comprehensible, several exemplary embodiments accompanied with drawings are described in detail below.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

FIG. 1A to FIG. 1K are cross-sectional views illustrating a manufacturing process of a HEMT device according to some embodiments of the invention.

DESCRIPTION OF THE EMBODIMENTS

The embodiments are described in detail below with reference to the accompanying drawings, but the embodiments are not intended to limit the scope of the invention. For the sake of easy understanding, the same components in the following description will be denoted by the same reference symbols. In addition, the drawings are for illustrative purposes only and are not drawn to the original dimensions. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A to FIG. 1K are cross-sectional views illustrating a manufacturing process of a HEMT device according to some embodiments of the invention.

Referring to FIG. 1A, a channel layer **100** is provided. In some embodiments, the channel layer **100** may be provided on a substrate (not shown). The substrate may be a semiconductor substrate such as a silicon substrate. In some embodiments, there may be a buffer layer (not shown) between the channel layer **100** and the substrate. The material of the buffer layer is, for example, aluminum nitride, aluminum gallium nitride, carbon-doped gallium nitride (C-doped GaN), or a combination thereof. The material of the channel layer **100** is, for example, gallium nitride. The method of forming the channel layer **100** is, for example, an epitaxial growth method.

A barrier layer **102** is formed on the channel layer **100**. The material of the barrier layer **102** is, for example, aluminum gallium nitride. The method of forming the barrier layer **102** is, for example, an epitaxial growth method.

A P-type gallium nitride material layer **104** may be formed on the barrier layer **102**. The method of forming the P-type gallium nitride material layer **104** is, for example, an epitaxial growth method.

A work function material layer **106** may be formed on the P-type gallium nitride material layer **104**. In addition, when the interface between the work function material layer **106** and the P-type gallium nitride material layer **104** forms an ohmic contact, the material of the work function material layer **106** may be tantalum (Ta), aluminum (Al), or titanium (Ti). In addition, when the interface between the work function material layer **106** and the P-type gallium nitride material layer **104** forms a Schottky contact, the material of the work function material layer **106** may be titanium nitride (TiN), tantalum nitride (Ta₂N₃), tungsten (W), nickel (Ni), or gold (Au). The method of forming the work function material layer **106** is, for example, an atomic layer deposition (ALD) method or a physical vapor deposition (PVD) method.

A hard mask material layer **108** may be formed on the work function material layer **106**. The material of the hard mask material layer **108** is, for example, silicon oxide. The method of forming the hard mask material layer **108** is, for example, a chemical vapor deposition (CVD) method.

A hard mask material layer **110** may be formed on hard mask material layer **108**. The material of the hard mask material layer **110** is, for example, silicon nitride. The method of forming the hard mask material layer **110** is, for example, a CVD method.

Referring to FIG. 1B, the hard mask material layer **110**, the hard mask material layer **108**, the work function material layer **106**, and the P-type gallium nitride material layer **104** may be patterned to form a hard mask layer **110a**, a hard mask layer **108a**, a work function layer **106a**, and a P-type gallium nitride layer **104a**. Therefore, the P-type gallium nitride layer **104a** may be formed on the barrier layer **102**, the work function layer **106a** may be formed on the P-type gallium nitride layer **104a**, the hard mask layer **108a** may be formed on the work function layer **106a**, and the hard mask layer **110a** may be formed on the hard mask layer **108a**. In some embodiments, the hard mask material layer **110**, the hard mask material layer **108**, the work function material layer **106**, and the P-type gallium nitride material layer **104** may be patterned by a lithography process and an etching process (e.g., dry etching process). Furthermore, a portion of the barrier layer **102** may be removed. Therefore, the thickness T1 of the barrier layer **102** located directly under the P-type gallium nitride layer **104a** is greater than the thickness T2 of the barrier layer **102** located on two sides of the P-type gallium nitride layer **104a**. In some embodiments, the portion of barrier layer **102** may be removed during patterning the hard mask material layer **110**, the hard mask material layer **108**, the work function material layer **106**, and the P-type gallium nitride material layer **104**.

Referring to FIG. 1C, the hard mask layer **110a** may be removed. The method of removing the hard mask layer **110a** is, for example, a wet etching method.

A spacer **112** may be formed on the sidewall SW1 of the P-type gallium nitride layer **104a** and on the barrier layer **102**. In addition, the spacer **112** may be further formed on the sidewall SW2 of the barrier layer **102**, the sidewall SW3 of the work function layer **106a**, and the sidewall SW4 of the hard mask layer **108a**. The spacer **112** may be a single-layer structure or a multilayer structure. The material of the spacer **112** is, for example, silicon oxide, silicon nitride, or a combination thereof. In some embodiments, the method of forming the spacer **112** may include the following steps, but the invention is not limited thereto. A spacer material layer (not shown) may be conformally formed on the barrier layer **102**, the P-type gallium nitride layer **104a**, the work function layer **106a**, and the hard mask layer **108a**. A dry etching process may be performed on the spacer material layer to form the spacer **112**.

Referring to FIG. 1D, a barrier layer **114** may be formed on the barrier layer **102**. The barrier layer **114** and the P-type gallium nitride layer **104a** may be separated from each other. In addition, the spacer **112** may be located between the barrier layer **114** and the P-type gallium nitride layer **104a**. In some embodiments, the method of forming the barrier layer **114** is, for example, an epitaxial growth method. In some embodiments, the method of forming the barrier layer **114** is, for example, a metal-organic chemical vapor deposition (MOCVD) method.

Referring to FIG. 1E, the height of the spacer **112** may be reduced, thereby exposing the sidewall SW3 of the work function layer **106a** and the sidewall SW4 of the hard mask

layer **108a**. The method of reducing the height of the spacer **112** is, for example, a dry etching method.

Referring to FIG. 1F, a lateral etching process may be performed on the work function layer **106a** to reduce the width of the work function layer **106a**. Therefore, the sidewall SW3 of the work function layer **106a** may not be aligned with the sidewall SW1 of the P-type gallium nitride layer **104a**. The lateral etching process is, for example, a dry etching process.

Referring to FIG. 1G, the hard mask layer **108a** may be removed. The method of removing the hard mask layer **108a** is, for example, a dry etching method. In some embodiments, after removing the hard mask layer **108a**, a pre-clean process may be performed.

A passivation layer **116** may be formed on the P-type gallium nitride layer **104a**, the spacer **112**, the barrier layer **114**, and the work function layer **106a**. The material of the passivation layer **116** is, for example, aluminum oxide or aluminum nitride. The method of forming the passivation layer **116** is, for example, an ALD method.

Referring to FIG. 1H, a dielectric layer **118** may be formed on the passivation layer **116**. The material of the dielectric layer **118** is, for example, silicon oxide. The method of forming the dielectric layer **118** is, for example, a CVD method.

An opening OP1 may be formed in the dielectric layer **118** and the passivation layer **116**. The opening OP1 may expose the work function layer **106a**. In some embodiments, the opening OP1 may be formed by removing a portion of the dielectric layer **118** and a portion of the passivation layer **116** by a lithography process and an etching process (e.g., dry etching process).

Referring to FIG. 1I, a contact **120** may be formed in the opening OP1. The contact **120** may be electrically connected to the P-type gallium nitride layer **104a**. In some embodiments, the contact **120** may be electrically connected to the work function layer **106a**, and the contact **120** may be electrically connected to the P-type gallium nitride layer **104a** by the work function layer **106a**. The material of the contact **120** is, for example, titanium, titanium nitride, aluminum copper alloy (AlCu), or a combination thereof. In some embodiments, the contact **120** may be formed by a deposition process, a lithography process, and an etching process.

Referring to FIG. 1J, a dielectric layer **122** may be formed on the dielectric layer **118** and the contact **120**. The material of the dielectric layer **122** is, for example, silicon oxide. The method of forming the dielectric layer **122** is, for example, a CVD method.

An opening OP2 and an opening OP3 may be formed in the dielectric layer **122**, the dielectric layer **118**, the passivation layer **116**, the barrier layer **114**, and the barrier layer **102**. The opening OP2 and the opening OP3 may respectively expose a portion of the channel layer **100**. In some embodiments, the opening OP2 and the opening OP3 may be formed by removing a portion of the dielectric layer **122**, a portion of the dielectric layer **118**, a portion of the passivation layer **116**, a portion of the barrier layer **114**, and a portion of the barrier layer **102** by a lithography process and an etching process (e.g., dry etching process).

Referring to FIG. 1K, a source electrode **124** and a drain electrode **126** may be formed in the opening OP2 and the opening OP3, respectively. The material of the source electrode **124** and the drain electrode **126** is, for example, titanium, titanium nitride, aluminum copper alloy, or a combination thereof. In some embodiments, the source

electrode **124** and the drain electrode **126** may be formed by a deposition process, a lithography process, and an etching process.

Hereinafter, the HEMT device **10** of the above embodiment is described with reference to FIG. 1K. Moreover, although the method for forming the HEMT device **10** is described by taking the above method as an example, the invention is not limited thereto.

Referring to FIG. 1K, the HEMT device **10** includes a channel layer **100**, a barrier layer **102**, and a P-type gallium nitride layer **104a**. The material of the channel layer **100** is, for example, gallium nitride.

The barrier layer **102** is disposed on the channel layer **100**. The material of the barrier layer **102** is, for example, aluminum gallium nitride. In some embodiments, the aluminum content in the aluminum gallium nitride may be 5 atomic % to 50 atomic %. The aluminum content in the aluminum gallium nitride can be used to adjust the concentration of the 2DEG. In some embodiments, when the material of the barrier layer **102** is aluminum gallium nitride, and the aluminum content in the aluminum gallium nitride is 5 atomic % to 50 atomic %, the concentration of the 2DEG of the HEMT device **10** can be increased, thereby improving the on-current (I_{on}) and the electrical performance of the HEMT device **10**.

The P-type gallium nitride layer **104a** is disposed on the barrier layer **102**. The P-type gallium nitride layer **104a** may be used as a gate. In some embodiments, the thickness of the P-type gallium nitride layer **104a** may be 80 nm to 100 nm.

In addition, the thickness T1 of the barrier layer **102** located directly under the P-type gallium nitride layer **104a** is greater than the thickness T2 of the barrier layer **102** located on two sides of the P-type gallium nitride layer **104a**. Therefore, the concentration of the 2DEG at the edge of the P-type gallium nitride layer **104a** can be reduced, thereby reducing the leakage current of the HEMT device **10** and improving the reliability of the HEMT device **10**. In some embodiments, the thickness T1 of the barrier layer **102** may be 14 nm to 24 nm, and the thickness T2 of the barrier layer **102** may be 2 nm to 22 nm. In addition, the upper surface S1 of the barrier layer **102** located directly under the P-type gallium nitride layer **104a** may be higher than the upper surface S2 of the barrier layer **102** located on two sides of the P-type gallium nitride layer **104a**.

The HEMT device **10** may further include at least one of a work function layer **106a**, a spacer **112**, a barrier layer **114**, a passivation layer **116**, a dielectric layer **118**, a contact **120**, a dielectric layer **122**, a source electrode **124**, and a drain electrode **126**. The work function layer **106a** is disposed on the P-type gallium nitride layer **104a**. In some embodiments, the width W2 of the work function layer **106a** may be smaller than the width W1 of the P-type gallium nitride layer **104a**, thereby reducing leakage paths. In some embodiments, the sidewall SW3 of the work function layer **106a** may not be aligned with the sidewall SW1 of the P-type gallium nitride layer **104a**. In addition, when the interface between the work function layer **106a** and the P-type gallium nitride layer **104a** forms an ohmic contact, the material of the work function layer **106a** may be tantalum (Ta), aluminum (Al), or titanium (Ti). Furthermore, when the interface between the work function layer **106a** and the P-type gallium nitride layer **104a** forms a Schottky contact, the material of the work function layer **106a** may be titanium nitride (TiN), tantalum nitride (TaN), tungsten (W), nickel (Ni), or gold (Au). In some embodiments, the thickness of the work function layer **106a** may be 30 nm to 50 nm.

The spacer **112** is disposed on the sidewall **SW1** of the P-type gallium nitride layer **104a** and on the barrier layer **102**. In some embodiments, the spacer **112** may be further disposed on the sidewall **SW2** of the barrier layer **102**. The spacer **112** may be a single-layer structure or a multilayer structure. The material of the spacer **112** is, for example, silicon oxide, silicon nitride, or a combination thereof. In some embodiments, the width of the spacer **112** may be 0.1 μm to 0.5 μm .

The barrier layer **114** is disposed on the barrier layer **102**. The barrier layer **114** and the P-type gallium nitride layer **104a** may be separated from each other. The spacer **112** may be located between the barrier layer **114** and the P-type gallium nitride layer **104a**. The barrier layer **114** can increase the concentration of the 2DEG under the barrier layer **114**, thereby improving the on-current (I_{on}) and the electrical performance of the HEMT device **10**. The material of the barrier layer **114** is, for example, aluminum gallium nitride. In some embodiments, the aluminum content in the aluminum gallium nitride may be 5 atomic % to 50 atomic %. The aluminum content in the aluminum gallium nitride can be used to adjust the concentration of the 2DEG. In some embodiments, when the material of the barrier layer **114** is aluminum gallium nitride, and the aluminum content in the aluminum gallium nitride is 5 atomic % to 50 atomic %, the concentration of the 2DEG of the HEMT device **10** can be increased, thereby improving the on-current (I_{on}) and the electrical performance of the HEMT device **10**. In some embodiments, the thickness of the barrier layer **114** may be 5 nm to 20 nm.

The passivation layer **116** is disposed on the P-type gallium nitride layer **104a**, the spacer **112**, the barrier layer **114**, and the work function layer **106a**. The passivation layer **116** can be used to block moisture. The material of the passivation layer **116** is, for example, aluminum oxide or aluminum nitride.

The dielectric layer **118** is disposed on the passivation layer **116**. The material of the dielectric layer **118** is, for example, silicon oxide. The contact **120** is electrically connected to the P-type gallium nitride layer **104a**. In some embodiments, the contact **120** may be electrically connected to the work function layer **106a**, and the contact **120** may be electrically connected to the P-type gallium nitride layer **104a** by the work function layer **106a**. In some embodiments, the contact **120** may pass through the dielectric layer **118** and the passivation layer **116** to be electrically connected to the work function layer **106a**. In some embodiments, a portion of the contact **120** may be disposed on the dielectric layer **118**. The material of the contact **120** is, for example, titanium, titanium nitride, aluminum copper alloy, or a combination thereof.

The dielectric layer **122** is disposed on the dielectric layer **118** and the contact **120**. The material of the dielectric layer **122** is, for example, silicon oxide. The source electrode **124** and the drain electrode **126** are disposed on two sides of the P-type gallium nitride layer **104a**. In some embodiments, the source electrode **124** and the drain electrode **126** may be connected to channel layer **100**, respectively. In some embodiments, the source electrode **124** and the drain electrode **126** may pass through the dielectric layer **122**, the dielectric layer **118**, the passivation layer **116**, the barrier layer **114**, and the barrier layer **102**, respectively, to be connected to channel layer **100**. In addition, a portion of the source electrode **124** and a portion of the drain electrode **126** may be disposed on the dielectric layer **122**. The material of

the source electrode **124** and the drain electrode **126** is, for example, titanium, titanium nitride, aluminum copper alloy, or a combination thereof.

Based on the above embodiments, in the HEMT device **10** and the manufacturing method thereof, the thickness **T1** of the barrier layer **102** located directly under the P-type gallium nitride layer **104a** is greater than the thickness **T2** of the barrier layer **102** located on two sides of the P-type gallium nitride layer **104a**. Therefore, the concentration of the 2DEG at the edge of the P-type gallium nitride layer **104a** can be reduced, thereby reducing the leakage current of the HEMT device **10** and improving the reliability of the HEMT device **10**.

In summary, in the HEMT device and the manufacturing method thereof of the aforementioned embodiments, since the thickness of the barrier layer located directly under the P-type gallium nitride layer is greater than the thickness of the barrier layer located on two sides of the P-type gallium nitride layer, the leakage current of the HEMT device can be reduced, and the reliability of the HEMT device can be improved.

Although the invention has been described with reference to the above embodiments, it will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit of the invention. Accordingly, the scope of the invention is defined by the attached claims not by the above detailed descriptions.

What is claimed is:

1. A high electron mobility transistor (HEMT) device, comprising:

a channel layer;

a first barrier layer disposed on the channel layer;

a P-type gallium nitride layer disposed on the first barrier layer;

a second barrier layer disposed on the first barrier layer, wherein the second barrier layer and the P-type gallium nitride layer are separated from each other;

a spacer disposed on a sidewall of the P-type gallium nitride layer and on the first barrier layer and located between the second barrier layer and the P-type gallium nitride layer; and

a passivation layer disposed on the P-type gallium nitride layer, the spacer, and the second barrier layer, wherein a first thickness of the first barrier layer located directly under the P-type gallium nitride layer is greater than a second thickness of the first barrier layer located on two sides of the P-type gallium nitride layer, and the passivation layer is in direct contact with the P-type gallium nitride layer and the second barrier layer.

2. The HEMT device according to claim 1, wherein the first thickness is 14 nm to 24 nm.

3. The HEMT device according to claim 2, wherein the second thickness is 2 nm to 22 nm.

4. The HEMT device according to claim 1, wherein a first upper surface of the first barrier layer located directly under the P-type gallium nitride layer is higher than a second upper surface of the first barrier layer located on two sides of the P-type gallium nitride layer.

5. The HEMT device according to claim 1, wherein a material of the first barrier layer comprises aluminum gallium nitride.

6. The HEMT device according to claim 5, wherein an aluminum content in the aluminum gallium nitride is 5 atomic % to 50 atomic %.

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7. The HEMT device according to claim 1, wherein a material of the second barrier layer comprises aluminum gallium nitride.

8. The HEMT device according to claim 7, wherein an aluminum content in the aluminum gallium nitride is 5 atomic % to 50 atomic %.

9. The HEMT device according to claim 1, wherein a width of the spacer is 0.1 μm to 0.5 μm .

10. The HEMT device according to claim 1, further comprising:

a work function layer disposed on the P-type gallium nitride layer.

11. The HEMT device according to claim 10, wherein a width of the work function layer is smaller than a width of the P-type gallium nitride layer.

12. A manufacturing method of a HEMT device, comprising:

providing a channel layer;

forming a first barrier layer on the channel layer;

forming a P-type gallium nitride layer on the first barrier layer;

forming a second barrier layer on the first barrier layer, wherein the second barrier layer and the P-type gallium nitride layer are separated from each other;

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forming a spacer on a sidewall of the P-type gallium nitride layer and on the first barrier layer, wherein the spacer is located between the second barrier layer and the P-type gallium nitride layer; and

forming a passivation layer on the P-type gallium nitride layer, the spacer, and the second barrier layer, wherein a first thickness of the first barrier layer located directly under the P-type gallium nitride layer is greater than a second thickness of the first barrier layer located on two sides of the P-type gallium nitride layer, and

the passivation layer is in direct contact with the P-type gallium nitride layer and the second barrier layer.

13. The manufacturing method of the HEMT device according to claim 12, further comprising:

forming a work function layer on the P-type gallium nitride layer.

14. The manufacturing method of the HEMT device according to claim 13, further comprising:

performing a lateral etching process on the work function layer to reduce a width of the work function layer.

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